

Scalability & Future Plan

Date	13 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited training dataset	Prediction accuracy may decrease for unseen or diverse applicant profiles.	High
2	Local deployment	The application is accessible only on the local machine and cannot be used remotely.	High
3	Basic input validation	Incorrect or incomplete user inputs may affect prediction quality.	Medium
4	No user account management	The application does not store user details or prediction history.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Access	Local Flask application	Deploy on Render or Railway for public access.
2	Dataset	Limited historical loan dataset	Train the model using a larger and more diverse dataset.
3	Performance	Single-user local execution	Optimize the application to support multiple concurrent users.
4	Prediction Accuracy	Current machine learning model	Experiment with additional algorithms and tune model parameters to improve performance.
5	User Experience	Basic prediction interface	Add prediction history, input validation, and improved UI features.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform (Render/Railway)	2 Months	Enable public access to the application.
Phase 2	Improve the machine learning model using additional training data	4 Months	Increase prediction accuracy and reliability.
Phase 3	Develop a mobile-friendly interface or Android application	6 Months	Improve accessibility across different devices.
Phase 4	Add user authentication and prediction history	8 Months	Enhance usability and allow users to track previous predictions.